

Medication	Anti-coagulation	Heparin	values	Start NMP: 0,2 mL (1000 units). Then every hour: 0,2 mL.	5000 IE/ml	
Arterial perfusate	Metabolites	Lactate (cLac)	<1,8 mmol/L after 150 min of NMP	Develops during anaerobe metabolism. A functioning liver clears lactate. Rising lactate levels correlate with liver dysfunction.		
		Glucose (cGluc)	>4,0 mmol/L	Decreases when a liver functions well due to metabolism. When the decrease is mild, the metabolism could be weak. Values can rise due to cellular leakage caused by IRI. Not the most important value to monitor.	Glucose 50% 2,775 mmol/mL	Strive 8-12mmol/L glucose. Use only 2-4 mL per correction.
	Electrolytes	Sodium (cNa ⁺)	135-145 mmol/L	Extracellular electrolyte Rises with a good metabolism due to the sodium/potassium channels. (Na out of the cell, K into the cell)	NaCl 5% 0,86 mmol/mL	Only administer if no NaBic is given, to increase sodium to 135mmol/L. Do not correct before T10.
		Potassium (cK ⁺)	3,5-5,0 mmol/L	Intracellular electrolyte. electrolyte Decreases with a good metabolism due to the sodium/potassium channels. (Na out of the cell, K into the cell)	KCl 74,6 mg/ml 1 mmol/mL	Only correct for hypokalemia. Do not over correct!
		Chloride (cCl ⁻)	>90 mmol/L	The Cystic fibrosis transmembrane conductance regulator (CFTR) channel facilitates transmembral chloride flow toward the bile. Anion exchange protein 2 (AE2) is a membrane transport protein that exchanges Cl ⁻ for HCO ₃ ⁻ in the bile. Alkali bile is dependent on Cl ⁻ values.	NaCl 5% 0,86 mmol/mL	
		Calcium (cCa ²⁺)	≥0,70 mmol/L	Ca ²⁺ is an important cofactor for bile production.	Calcium-gluconate 0,225 mmol/mL	Add 10 mL at the start of COR, afterwards usually another 10 mL when the liver is warm. Add per 5 mL.
	Blood gas	pH	COR: <7.30 NMP: 7,35-7,45	Two factors influence the pH: 1) Metabolism: more acidic during anaerobe metabolism due to lactate levels. At the start of NMP there is a pH drop due to lactate. There could be a shortage of HCO ₃ ⁻ . A mild acidic environment could be protective against IRI. 2) Gas exchange: CO ₂ is acidic, a higher airflow results in a higher clearance of CO ₂ .	NaHCO ₃ 8.4% 1 mmol/ml	
		pCO ₂	4.5-6.0 kPa	An acidic product of metabolic activity and a substrate of HCO ₃ ⁻ . You can get rid of it by increasing the gas flow.	↓ gas flow = ↓ pH ↑ gas flow = ↑ pH	Be really careful with your airflow during COR, since there is almost no CO ₂ production. Only turn up the gasflow if the liver starts producing CO ₂ . Maintain a stable pCO ₂ during the perfusion, adjust when necessary.
		pO ₂	During COR & NMP: 10-12 kPa	Needed for aerobic metabolism. Hypoxia is toxic due to reactive oxygen species.	Gas flow (ml/min) FIO ₂ (%)	Do not let the go below 10kPa. Increase FIO ₂ with 1 to 2 %